IP ADDRESSING & SUBNETTING

TASK-1 List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

IP Class	Address Range (First Octet)	Purpose	Real-World Example
Class A	1–126	Large networks	A large organization using a private network such as 10.0.0.1
Class B	128–191	Medium-sized networks	A college network using 172.16.0.1
Class C	192–223	Small networks	A home router using 192.168.1.1
Class D	224–239	Multicast communication	Streaming video to multiple devices using a multicast IP address
Class E	240–255	Experimental or reserved purposes	Networking research and testing

TASK-2 Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses Hint Show your working steps for each calculation.

Given

- IP address: 192.168.10.45
- Subnet mask: 255.255.255.0

1. Network Address

Working: IP address AND subnet mask.

- IP: 192.168.10.45
- Mask: 255.255.255.0

Network address: 192.168.10.0

2. Broadcast Address

Broadcast address: 192.168.10.255

3. Valid Host Address Range

- First valid host = Network address + 1
- Last valid host = Broadcast address − 1

Valid host range: 192.168.10.1 – 192.168.10.254

Item	Result
Network address	192.168.10.0
Broadcast address	192.168.10.255
First valid host	192.168.10.1
Last valid host	192.168.10.254
Total valid host addresses	254

Formula:

Valid hosts = 2^h - 2 = 2^8 - 2 = 254 where h=8 host bits.

TASK-3 Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Calculation

- Original network: 10.0.0.0/24
- Required subnets: 4
- Borrowed bits: 2 because 2^2 = 4
- New prefix: 24 + 2 = 26
- Hosts per subnet: 2^6 - 2 = 62 usable addresses

TASK-4 You are given the following scenario: A college Wi Fi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint: Show your calculation steps and final answers.

Given

- IP network: 172.16.0.0/16
- Required subnets: 20

Step 1: Calculate the borrowed bits

Formula:

- 2^4 = 16 — not enough
- 2^5 = 32— sufficient

Therefore, borrow 5 bits from the host portion.

Step 2: Calculate the new subnet mask

New prefix = 16 + 5 = 21

The new subnet mask is 255.255.248.0.

Step 3: Calculate hosts per subnet

Remaining host bits:

32 - 21 = 11

Usable hosts per subnet:

2^(11) - 2 = 2048 - 2 = 2046

Final Answer

Item	Answer
Original network	172.16.0.0/16
Required subnets	At least 20
Bits borrowed	5
Total subnets created	32

New subnet mask	255.255.248.0
New prefix	21
Host bits remaining	11

TASK-5 A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

The IP address 192.168.1.300 is invalid because each octet in an IPv4 address must be between 0 and 255. The last octet, 300, exceeds this limit.

A valid IP address in the same subnet could be 192.168.1.100, assuming it is not already in use and is within the subnet's valid host range.